Eenheid 6.3

Die Puntproduk

P 800 – 803

Definisie: Laat $\bar{a} = \langle a_1, a_2 \rangle$ en
$\bar{b} = \langle b_1, b_2 \rangle$ in $\mathbb{R}^2$

Ons definieer die puntproduk van $\bar{a}$ en $\bar{b}$ as

$$\bar{a} \cdot \bar{b} = \langle a_1, a_2 \rangle \cdot \langle b_1, b_2 \rangle$$
$$= a_1 b_1 + a_2 b_2$$

Antwoord is 'n Getal !!!

In $\mathbb{R}^3$ is $\bar{a} \cdot \bar{b} = \langle a_1, a_2, a_3 \rangle \cdot \langle b_1, b_2, b_3 \rangle$

$= a_1 b_1 + a_2 b_2 + a_3 b_3$

antwoord is 'n getal !!

VOORBEELDE:

1) $\langle -2, 3 \rangle \cdot \langle 4, -5 \rangle = -8 + (-15) = -8 - 15 = -23$

2) $(8\bar{i} - 2\bar{j} + 3\bar{k}) \cdot (2\bar{i} + 4\bar{k}) = \langle 8, -2, 3 \rangle \cdot \langle 2, 0, 4 \rangle$

$= 16 + 0 + 12$

$= 28$

$\boxed{\bar{i} = \hat{i}}$

3) $\langle 2, 3, -1 \rangle \cdot \bar{j} = \langle 2, 3, -1 \rangle \cdot \langle 0, 1, 0 \rangle$

$= 2 \times 0 + 3 \times 1 + -1 \times 0 = 0 + 3 + 0$

$= 3$

Eienskappe

Laat $\bar{a}, \bar{b}, \bar{c} \in \mathbb{R}^n$ en c is skalaar ($c \in \mathbb{R}$),

dan

1) $\bar{a} \cdot \bar{a} = |\bar{a}|^2$

2) $\bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{a}$ (kommutatief)

3) $\bar{a} \cdot (\bar{b} + \bar{c}) = \bar{a} \cdot \bar{b} + \bar{a} \cdot \bar{c}$ (Distributiewe)

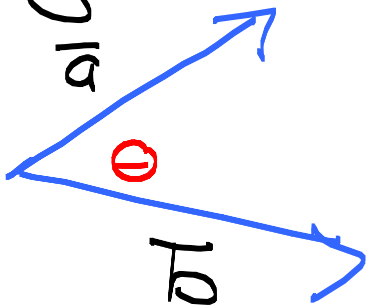
4) $(c\bar{a}) \cdot \bar{b} = c(\bar{a} \cdot b)$

$\bar{a} \cdot (c\bar{b}) = c(\bar{a} \cdot \bar{b})$

5) $\bar{o} \cdot \bar{a} = 0$ (antw is die getal 0)

Definisie

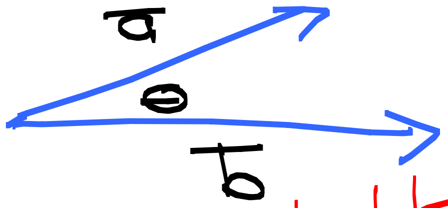
Laat $\bar{a}$ en $\bar{b}$ vektore wees in $\mathbb{R}^n$.
Die hoek θ tussen $\bar{a}$ en $\bar{b}$ word gedefinieer as die hoek tussen die voorstelling wat by die oorsprong begin en $0 \le \theta \le \pi$



Let op: As $\bar{a}$ en $\bar{b}$ parallel is dan is $\theta = 0$ of $\theta = \pi$

Stelling

Laat θ die hoek tussen $\bar{a}$ en $\bar{b}$ wees,



dan is $\bar{a} \cdot \bar{b} = |\bar{a}||\bar{b}| \cos\theta$

As $\bar{a} \neq \bar{0}$ en $\bar{b} \neq \bar{0}$ dan

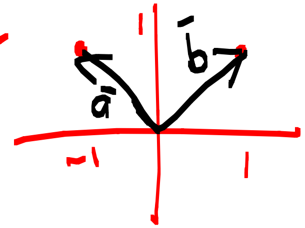
$$\cos\theta = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}||\bar{b}|}$$

$$\boxed{\begin{array}{l} \text{Sê: } \cos\theta = 5 \\ \theta = \text{bg}\cos 5 \end{array}}$$

Voorbeelde ① Bepaal die hoek tussen

$\bar{a} = \langle -1, 1 \rangle$ en $\bar{b} = \langle 1, 1 \rangle$

$$\cos\theta = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}||\bar{b}|} = \frac{-1+1}{(\sqrt{2})(\sqrt{2})} = \frac{0}{2} = 0 \Rightarrow \theta = \frac{\pi}{2} \checkmark$$



$$\left(\theta = \text{bg}\cos 0\right) = \pi/2$$

ONTHOU: $0 \leq \theta \leq \pi$

② Bepaal die hoek tussen $\bar{a} = \langle 1,-2,3\rangle$ en $\bar{b} = \langle -3,2,1\rangle$

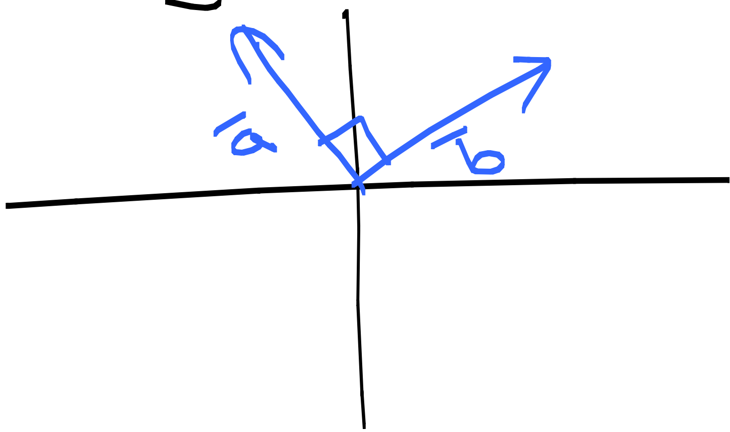
$$\cos\theta = \frac{\langle 1,-2,3\rangle \cdot \langle -3,2,1\rangle}{(\sqrt{14})(\sqrt{14})} = \frac{-4}{14} = -\frac{2}{7}$$

$$\cos\theta = -\frac{2}{7} \Rightarrow \theta = \text{bgcos}\left(-\frac{2}{7}\right)$$

Definisie Twee nie-nul vektore $\bar{a}$ en $\bar{b}$ is ortogonaal (loodreg) as die hoek tussen die twee vektore $\theta = \frac{\pi}{2}$ is

Bv. $\bar{a} = \langle -1,3\rangle$ en $\bar{b} = \langle 3,1\rangle$

$\bar{a}\cdot\bar{b} = (-1)(3) + (3)(1) = 0$



As $\cos\theta = \dfrac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$ en as $\theta = \frac{\pi}{2}$

Dan $\bar{a}\cdot\bar{b} = \cos\theta\,|\bar{a}||\bar{b}| = 0$

Dus: Twee nie-nul vektore is ortogonaal as en slegs as $\bar{a} \cdot \bar{b} = 0$

Bv. Bepaal k sodat $\bar{a} = \langle -1, 3, k \rangle$ en $\bar{b} = \langle 2, k, 3 \rangle$ ortogonaal is.

getal

$$\bar{a} \perp \bar{b} \Rightarrow \langle -1, 3, k \rangle \cdot \langle 2, k, 3 \rangle = 0$$

$$\Rightarrow -2 + 3k + 3k = 0$$

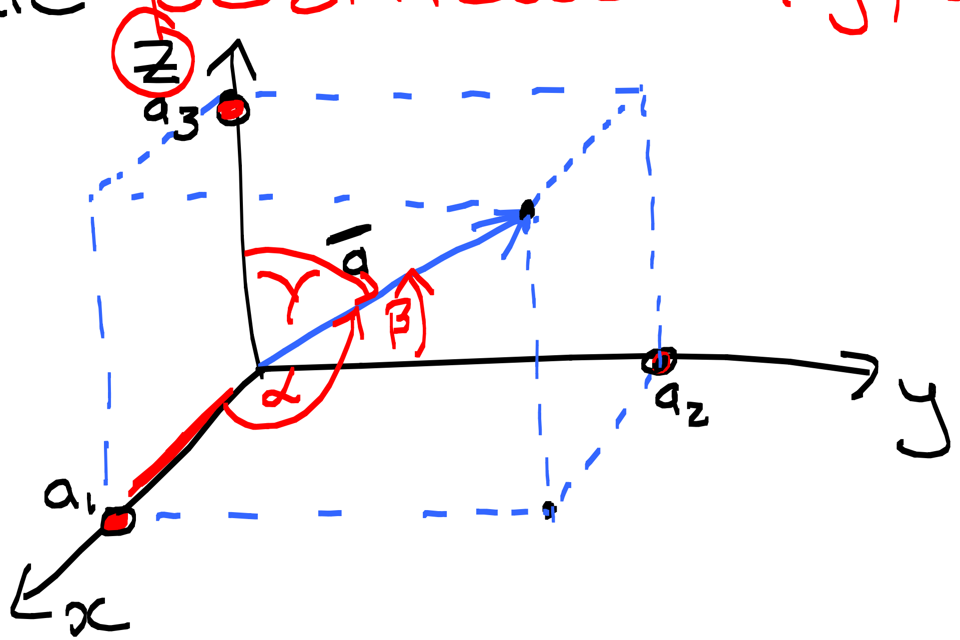
$$\Rightarrow 6k = 2$$

$$\Rightarrow k = \frac{1}{3}$$

Rigtingshoeke en Rigtingskosinusse

Definisie :

Rigtingshoeke van $\bar{a}$: is die hoeke α, β, γ in $[0, \pi]$ wat $\bar{a}$ maak met die positiewe x, y, z asse



$$\bar{a} = \langle a_1, a_2, a_3 \rangle$$

P803

Rigtingskosinusse : kosinusse van die rigtingshoeke

$\cos\alpha$, $\cos\beta$, $\cos\gamma$

Ons weet dat die hoek tussen $\bar{a}$ en $\bar{b}$ gegee word deur $\cos\theta = \dfrac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$ (algemeen

Nou: $\cos\alpha = \dfrac{\bar{a}\cdot\bar{i}}{|\bar{a}||\bar{i}|} = \dfrac{\langle a_1, a_2, a_3\rangle \cdot \langle 1,0,0\rangle}{|\bar{a}| \times 1}$ (Vervang $\bar{b}$ met $\bar{i}$)

$= \dfrac{a_1}{|\bar{a}|}$

Netso $\cos\beta = \dfrac{a_2}{|\bar{a}|}$ $(\bar{j} = \langle 0,1,0\rangle)$

$\cos\gamma = \dfrac{a_3}{|\bar{a}|}$ $(\bar{k} = \langle 0,0,1\rangle)$

Dan is $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$

$$= \frac{a_1^2}{|\bar{a}|^2} + \frac{a_2^2}{|\bar{a}|^2} + \frac{a_3^2}{|\bar{a}|^2}$$

eienskap

$\bar{a} \cdot \bar{a} = |\bar{a}|^2$

$$= \frac{\langle a_1, a_2, a_3 \rangle \cdot \langle a_1, a_2, a_3 \rangle}{|\bar{a}|^2} = \frac{|\bar{a}|^2}{|\bar{a}|^2} = 1$$

Dus: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Ook $\bar{a} = \langle a_1, a_2, a_3 \rangle$ $\left(\cos\alpha = \frac{a_1}{|\bar{a}|}\right)$

$$= \langle |\bar{a}|\cos\alpha, |\bar{a}|\cos\beta, |\bar{a}|\cos\gamma \rangle$$

$$= |\bar{a}| \langle \cos\alpha, \cos\beta, \cos\gamma \rangle$$

$$\Rightarrow \frac{\bar{a}}{|\bar{a}|} = \langle \cos\alpha, \cos\beta, \cos\gamma \rangle$$

- Die Rigtingskosinusse is komponente van die eenheidsvektor in die rigting van $\bar{a}$.

VBD Bepaal die rigtingskosinusse van

$\bar{v} = 2\bar{i} - 4\bar{j} + 4\bar{k}$

asook die rigtingshoeke.

$|\bar{v}| = \sqrt{2^2 + (-4)^2 + 4^2} = \sqrt{36} = 6$

Eenheidsvektor in die rigting van $\bar{v}$ is

$\frac{\langle 2, -4, 4 \rangle}{|\bar{v}|} = \langle \frac{2}{6}, \frac{-4}{6}, \frac{4}{6} \rangle = \frac{1}{3}\bar{i} - \frac{2}{3}\bar{j} + \frac{2}{3}\bar{k}$

Rigtingskosinusse: $\cos\alpha = \frac{1}{3}$

$\cos\beta = -\frac{2}{3}$

$\cos\gamma = \frac{2}{3}$

Rigtingshoeke: $\alpha = \text{bgcos}\left(\frac{1}{3}\right)$ $\gamma = \text{bgcos}\left(\frac{2}{3}\right)$

$\beta = \text{bgcos}\left(-\frac{2}{3}\right)$